VISHAL

vishal05.official@gmail.com LinkedIn GitHub Portfolio

Education

Dr. Akhilesh Das Gupta Institute of Professional Studies (ADGIPS), GGSIPU **Present – 2028**
B.Tech – Artificial Intelligence & Data Science

G.B. Pant Institute of Technology **2020 – 2023**
Diploma – Mechanical Engineering

Tinu Public School **2019 – 2020**
Class X (CBSE Board)

Experience

Design Consultant (Intern) **Sep 2023 – Feb 2024**
WIKA Instruments Pvt. Ltd.

- Gained hands-on experience with industrial product development and engineering workflows by helping with design and engineering tasks. Worked together with the design team to comprehend product design procedures, precise engineering needs, and technical specifications, developing a solid grasp of industry best practices and professional design standards while contributing to technical documentation.

Industrial Trainee **May 2023 – Jun 2023**
Railway Coach Care Center (Delhi)

- Received hands-on experience with railway coach repair and inspection procedures while completing industrial training at the Railway Coach Care Center, working alongside RTR machine inspections, taking part in train roll-in and roll-out operations, and gaining a solid understanding of safety procedures, quality inspection methods, and preventative maintenance techniques in an industrial setting.

Projects

RepoAI OS **Jul 2026**

JavaScript, Node.js, MongoDB, OpenAI API, GitHub OAuth, MCP

- Built an engineering-intelligence platform to ingest local and public GitHub repositories and analyze code structure, languages, imports, functions, endpoints, and architecture.
- Implemented evidence-backed code search and optional OpenAI-powered repository Q&A grounded in retrieved source-code excerpts.
- Developed Git change-impact analysis to identify affected files, endpoints, and potential test gaps.
- Added automated security scanning for exposed credentials, injection risks, dynamic execution, JWT verification, retries, and dependency issues.
- Generated repository overview, architecture, API, and test-intelligence documentation; added GitHub OAuth, MongoDB persistence, and scoped MCP access tokens.

Roadscapes VLM Comparison under Day and Night Conditions **Oct 2025**

Python, Pandas, NumPy, Matplotlib, Hugging Face Transformers

- Built a benchmarking pipeline to evaluate four Vision-Language Models (Phi-3, Qwen2-VL, LLaVA, and PaliGemma) on the Roadscapes dataset.
- Evaluated Object Counting, Object Description, and Surrounding Description under day and night conditions.
- Generated automated comparison tables and visualizations for category-wise and overall accuracy.
- Qwen2-VL achieved the highest overall accuracy (49.33%), followed by Phi-3 (45.83%), LLaVA (42.83%), and PaliGemma (40.00%).

Skills

Languages: Java, Python, C, JavaScript
Core CS: Object-Oriented Programming, Data Structures & Algorithms
Web & Backend: HTML, CSS, Node.js, MongoDB, REST APIs
ML / Libraries: Pandas, NumPy, Matplotlib, Hugging Face Transformers, OpenAI API
Tools: Git, GitHub

Certifications

- NSIC AutoCAD Certification
- Generative AI Course powered by Google Cloud